

Arun Madhusudhanan

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EDUCATION

Master of Science in Robotics, Computer Science Concentration

Sep 2022 - Dec 2024

Northeastern University, Boston, MA

GPA: 4.0/4.0

Bachelor of Technology in Mechanical Engineering, NIT Calicut, India

Jul 2014 - May 2018

TECHNICAL SKILLS

Programming Languages: C++, Python

Libraries / Frameworks: PyTorch, OpenCV, Open3D, TensorRT, ONNX, OpenVINO, vLLM, Azure ML, Wandb, Nerfstudio

Software Tools: Docker, Git, Ubuntu, ROS, FastAPI, MATLAB, OriginPro, SolidWorks

WORK EXPERIENCE

Robotics & AI Software Engineering Intern

Sep 2025 – Nov 2025

forREAL (Startup), Danvers, MA

- Developed a **Gaussian Splatting pipeline** for generating high-quality 3D virtual tours for apartment listings, and prototyped **monocular depth guided supervision** to improve depth consistency of splats.
- Created an automated **Vision-Language Model pipeline** to select optimal apartment listing images from 3,000+ frames, achieving a **3× reduction in operational costs** through efficient local model serving with **vLLM**.
- Fine-tuned **Qwen3-4B** using **LoRA** to extract **normalized apartment search constraints and preferences** from conversational AI chatbot queries; deployed **locally via vLLM**.

Machine Learning Research co-op

Jul 2023 - Dec 2023

Festo Corporation, Marlborough, MA

- Developed a robust machine learning pipeline to predict the output parameters of a high-precision liquid dosing unit achieving **<2.5% error**; the device with integrated ML model is currently under **patent application**.
- Owned the **end-to-end ML lifecycle** using **Azure ML**, including data ingestion, labeling, feature extraction, model training and deployment via managed online endpoints.
- Optimized software modules for sensor activation, hardware drivers, and data storage by refactoring code, fixing bugs, and implementing multi-threading, resulting in a significant reduction in system latency.

Wells Engineer

Jul 2018 - Jun 2022

ExxonMobil, Bengaluru, India

- Led beta testing** of company-wide tubular design software, ensuring robustness and reliability before release to users.
- Stewarded and improved the tubular design workflow for business divisions across the world in accordance with industry standard API 5C5, resulting in **\$230k** immediate savings and long-term synergistic benefits.

PROJECTS

LoRA Fine-Tuning & Inference of Quantized Qwen2-VL Models [[Code](#)]

Jun 2025

- Fine-tuned two **4-bit** variants of the Qwen2-VL-2B on a LaTeX OCR dataset: (1) **GPTQ**-quantized model using **LoRA** with Hugging Face **TRL**, and (2) **NF4**-quantized model using **QLoRA** via **Unsloth**.
- Achieved over **2×** improvement in **BLEU** score and **4.5×** reduction in **Levenshtein Edit Distance** via LoRA fine-tuning.
- Enabled high-throughput, low-latency inference using **vLLM** to serve LoRA adapters without merging, benchmarked GPTQ vs QLoRA pipelines on **throughput**, **TTFT**, **TPOT**, and **ITL**, shared insights through detailed **medium articles**.
- Contributed to **open-source**: fixed model layer naming bugs in **vllm-project/llm-compressor** and **huggingface/optimum** and added support for two new datasets for multimodal benchmarking in **vllm-project/vllm**.

Open Vocabulary 3D Scene Understanding [[Report](#)]

Dec 2024 - Apr 2025

- Designed a **multimodal** 3D scene understanding pipeline enabling robots to interpret complex natural language, image, or audio commands for tasks like grasping and navigation.
- Leveraged vision foundation models (**RAM**, **Grounding DINO**, **SAM**, **CLIP**) for object detection, segmentation, and semantic feature extraction from 2D images, and projected results onto point clouds for **3D semantic segmentation**.
- Incorporated **scene graphs** with object-room hierarchies and used **Qwen2.5-7B-Instruct** for user query understanding, improving object retrieval accuracy by **95%** on the **Replica dataset**.
- Integrated **GraspNet** for open-vocabulary robot grasping in **PyBullet**, enhancing the pipeline with **SoM prompting** and a **VLM (LLaMA 4 Scout)** for improved reasoning and spatial understanding. [[Code](#)]

Other Projects

Jan 2023 - Aug 2025

- Implemented distributed training for GPT-2 using **Data Parallelism (FSDP)**, **Pipeline Parallelism**, and **Tensor Parallelism** in PyTorch on multi-GPU setup. [[Code](#)]
- Achieved **3–7× speedups** for **CLIP**, **CoCa**, and **YOLOv8** (with NMS TRT plugin) by **TensorRT** conversion. [[Code](#)]
- Optimized YOLOv8 CPU inference using **ONNX** and **OpenVINO** with **INT8** quantization, achieving **3×** speedup. [[Code](#)]